Concepção e Autoria Multimédia

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Game



- **A game *is a goal-directed and competitive activity conducted within a framework of agreed rules***.
- The **rules** establish what a player can or cannot do, and what the **behavioral consequences of actions** may be within the world of the game.
- Learning to play a game, making progress within a game, and, with persistence and basic ability, eventually completing or winning a game are a matter of **learning how to interact within the game system and its rules** in a way that supports progress.
- **Gameplay** can have many forms for a particular game, capturing different playing styles, tactics and approaches to progressing through the game and (perhaps) eventually winning. In general, it is a **particular way of thinking about the game state from the perspective of a player**, together with a pattern of repetitive perceptual, cognitive, and motor operations.

Game concepts



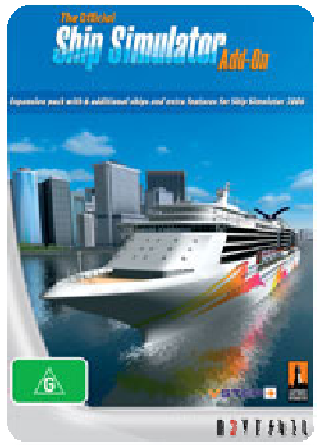
- Some examples of gameplay include:
  - *Action games*: shoot while being hit, strafe to hiding spot, take health, repeat
  - *RPGs*: send fast character to lure enemy from group, all characters kill enemy, take health, repeat
  - *Strategy Games*: order peasants, send to work, order soldiers, send to perimeters, repeat while slowly expanding the perimeters (up to the point of catastrophic win/lose);
  - *In General*: overcome barrier, save if successful, reload and retry if unsuccessful
- **Narrative is an experience that is structured in time**. Different structures then represent different forms of narrative, and a narrative is an experience manifesting a specific narrative structure.
- A very common narrative structure used in computer games, borrowed from film scriptwriting, is the **three-act restorative structure**. The three act restorative structure has a beginning (the first act) in which a conflict is established, followed by the playing out of the implications of the conflict (the second act), and completed by the final resolution of the conflict (the third act).

Game concepts



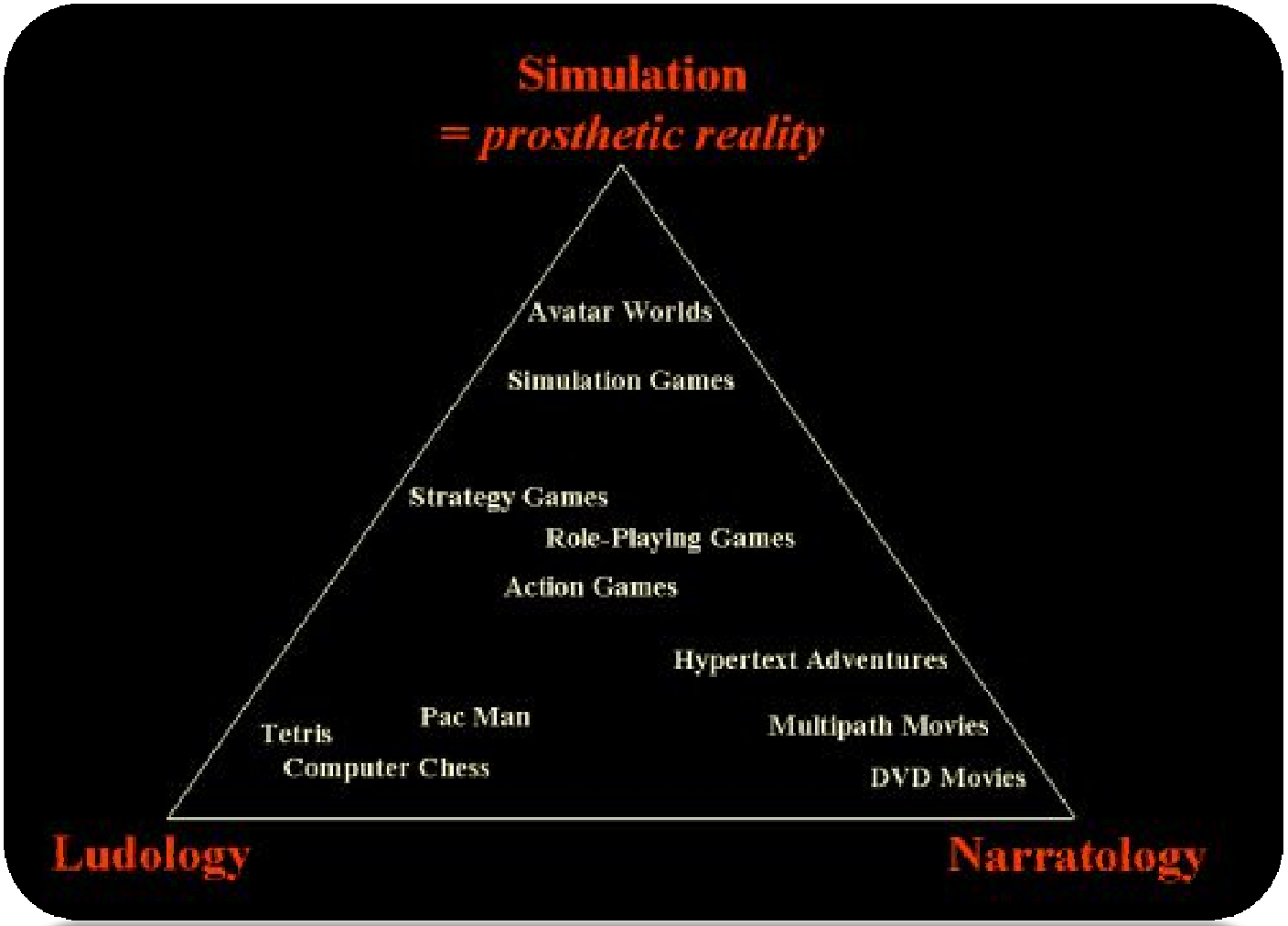
- The **game level** is designed for the pursuit of a goal, that of the player reaching the end of the level, that serves the purpose of progressing the player through the second act of the higher level three-act structure of the game narrative.
- **Conflict is achieved by resistance to the player achieving that goal**, in the form of opposing enemies, puzzles, barriers, and the like. The sense of level-specific conflict can be enhanced by increasing difficulty through a level, or by an internal dramatic structure that emphasizes the point of completing the level, such as the defeat of a level boss, the big barrier creature at the end of the level.
- **Simulation** is *a representation of the function, operation or features of one process or system through the use of another*.
- Hence a simulation may involve no specific repetitive and goal-oriented activities (there may be no obvious end state, other than the player getting bored), and no specific predefined patterns in time. Time patterns emerge over the course of running a simulation, and can be completely different for different runs.

Game concepts



- During competitive play, there is an obvious goal. But many **simulation games will allow to continue playing after all of the enemies are defeated**. Until resources run out, these games may then chug along indefinitely simulating a simple economic system.
- There is no more gameplay by a strict ludic definition, and the narrative after winning has no interesting temporal (dramatic) structure.
- Simulations like flight simulators are often interesting from the perspective of **skill development**; they are not interesting as games or stories, but for understanding how a particular system functions in different circumstances.

Taxonomy (Craig Lindley)



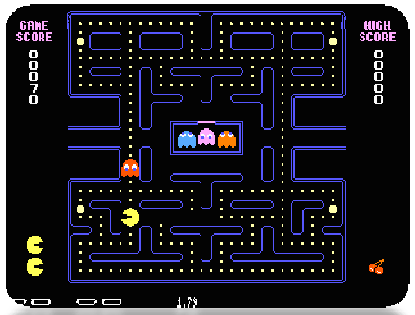
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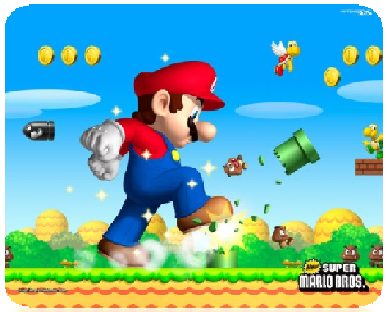


Game Taxonomy

- **Action:** An action game requires players to use quick reflexes and timing to overcome obstacles. It is perhaps the most basic of gaming genres, and certainly one of the broadest. Action games tend to have gameplay with emphasis on combat. There are many genres of action games, such as fighting games and first-person shooters.
  - **Beat 'em up and hack and slash games** have an emphasis on one-on-many close quarters combat, beating large numbers of computer-controlled enemies. Gameplay involves the player fighting through a series of increasingly difficult levels. The sole distinction between these two genres are that beat 'em ups feature hand-to-hand combat, and hack and slash games feature melee weaponry, particularly bladed weapons.
  - **Fighting games** emphasize one-on-one combat between two characters, one of which may be computer controlled. These games are usually played by linking together long chains of button presses on the controller to use physical attacks to fight. Many of the movements employed by the characters are usually dramatic and occasionally physically impossible.

Game Taxonomy





- **Maze games** have a playing field which is entirely a maze, which players must navigate. Quick thinking and fast reaction times are encouraged by the use of a timer, monsters obstructing the player's way, or multiple players racing to the finish. The most famous game of this genre is *Pac-Man*.
- **Pinball games** are essentially virtual pinball tables, designed to replicate the look and feel of a common pinball table. Most pinball games feature the same gameplay style, where the player controls a right and left flipper, and tries to make the ball hit various parts of the playfield to gather up points.
- **Platform games** (*platformers*) are a subgenre of action game. These games involve traveling between platforms by jumping (very occasionally other means are substituted for jumping, like swinging or bouncing, but these are considered variations on the same mechanic). Other traditional elements include running and climbing ladders and ledges. They are most often associated with iconic cartoon mascots like Donkey Kong, Sonic the Hedgehog, Mario, and Rayman, though platform games may have any theme.

Game Taxonomy



- **Shooter:** A shooter game focuses primarily on combat involving projectile weapons, such as guns and missiles. They can be divided into first-person and third-person shooters, depending on the camera perspective. Some first-person shooters use light gun technology.
  - **First-person shooter video games**, commonly known as FPSs, emphasize shooting and combat from the perspective of the character controlled by the player. This perspective is meant to give the player the feeling of "being there", and allows the player to focus on aiming. Most FPSs are very fast-paced and require quick reflexes on high difficulty levels. *Wolfenstein 3D* was the first widely known FPS, and *Doom* was the first major breakthrough in graphics;
  - **Massively multiplayer online first person shooter games** (MMOFPS) are a genre of massively multiplayer online games that combines first-person shooter gameplay with a virtual world in which a large number of players may interact over the Internet. Whereas standard FPS games limit the number of players able to compete in a multiplayer match (generally the maximum is 64), hundreds of players can battle each other on the same server in an MMOFPS.

Manga Fighter





Game Taxonomy

- A **shoot 'em up** (or *shmup* for short) is a genre of shooter game in which the player controls a character or vehicle and shoots large numbers of enemies, while dodging incoming projectiles. Games in this genre call for fast reactions and memorization of enemy patterns.
- **Third-person shooter video games**, known as TPSs or 3PSs, emphasize shooting and combat from a camera perspective in which the player character is seen at a distance. This perspective gives the player a wider view of their surroundings as opposed to the limited viewpoint of first-person shooters.
- **Tactical shooters** are variations on the first- and third-person shooter genre, which focus on realism and emphasize tactical play such as planning and teamwork (for example, co-ordination and specialised roles). In single player modes, the player commands a squad of AI controlled characters in addition to their own; in multi-player modes, players must work in teams to win the game. Winning is likely to be dependent on capturing an objective of some sort rather than gaining the most kills.

Game Taxonomy



- **Adventure games** were some of the earliest games created, beginning with the text adventure *Colossal Cave Adventure* in the 1970s. That game was originally titled simply "Adventure," and is the namesake of the genre. Over time, graphics have been introduced to the genre and the interface has evolved.
- Adventure describes a manner of gameplay without reflex challenges or action. They normally require the **player to solve various puzzles by interacting with people or the environment**, most often in a non-confrontational way.
- Because they put little pressure on the player in the form of action-based challenges or time constraints, adventure games have had the unique ability to appeal to people who do not normally play video games. The genre peaked in popularity with the 1993 release of *Myst*, the best-selling PC game of all time up to that point.
- A **visual novel** is an adventure game game featuring mostly static graphics, usually with anime-style art. As the name might suggest, they resemble mixed-media novels or tableau vivant stage plays.

Game Taxonomy





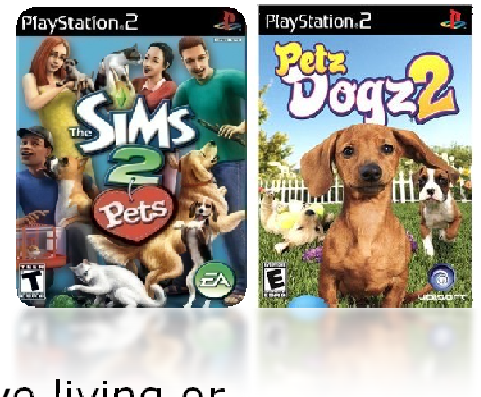
- **Action-adventure games** combine elements of their two component genres, typically featuring long-term obstacles that must be overcome using a tool or item as leverage (which is collected earlier), as well as many smaller obstacles almost constantly in the way, that require elements of action games to overcome. Action-adventure games tend to focus on exploration and usually involve item gathering, simple puzzle solving, and combat.
- **Stealth games** are a somewhat recent sub-genre, sometimes referred to as "sneakers" or "creepers" to contrast with the action-oriented "shooter" sub-genre. These games tend to emphasize subterfuge and precision strikes over the more overt mayhem of shooters.
- **Survival horror games** focus on fear and attempt to scare the player via traditional horror fiction elements such as atmospherics, death, the undead, blood and gore. One crucial gameplay element in many of these games is the low quantity of ammunition, or number of breakable mêlée weapons.

Game Taxonomy



- **Construction and management simulation** (CMSs) are a type of simulation game which task players to build, expand or manage fictional communities or projects with limited resources.
- In **city-building games** the player acts as overall planner or leader to meet the needs and wants of game characters by initiating structures for food, shelter, health, spiritual care, economic growth, etc. Success is achieved when the city budget makes a growing profit and citizens experience an upgraded lifestyle in housing, health, and goods.
- **Business simulation games** attempt to simulate an economy or business, with the player controlling the economy of the game.
- Unlike other genres of games, **god games** often do not have a set goal that allows a player to win the game. The focus of a god game tends to be control over the lives of people, anywhere from micromanaging a family to overseeing the rise of a civilization.
- A **government simulation game** (or "political game") involves the simulation of the policies, government or politics of a country, but typically excludes warfare.

Game Taxonomy



- **Life simulation games** (or Artificial Life Games) involve living or controlling one or more artificial lives. A life simulation game can revolve around individuals and relationships, or it could be a simulation of an ecosystem.
- **Biological simulations** may allow the player to experiment with genetics, survival or ecosystems, often in the form of an educational package.
- **Pet-raising simulations** (or digital pets) focus more on the relationship between the player and one or few life forms. They are often more limited in scope than other biological simulations. This includes popular examples of virtual pets such as the *Petz* series, and *Nintendogs*.
- Social simulation games base their gameplay on the social interaction between multiple artificial lives. The most famous example from this genre is Will Wright's *The Sims*.

Game Taxonomy

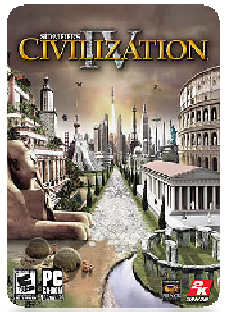




- **Role-playing games** (CRPGs or simply RPGs) draw their gameplay from traditional role-playing games like *Dungeons & Dragons*. Most cast the player in the role of one or more "adventurers" who specialize in specific skill sets (such as combat or casting magic spells) while progressing through a predetermined storyline.
- Gameplay elements strongly associated with RPGs, such as statistical character development through the acquisition of experience points, have been widely adapted to other genres such as action-adventure games.
- The **action role-playing game** or action RPG is a type of role-playing game which incorporates elements from action games or action-adventure games.
- **Massively multiplayer online role-playing games**, or MMORPGs, emerged in the mid to late 1990s as a commercial, graphical variant of text-based MUDs, which had existed since 1978. By and large, MMORPGs feature the usual RPG objectives of completing quests and strengthening one's player character, but involve up to hundreds of players interacting with each other on the same persistent world in real-time.

Shattered Union





Game Taxonomy

- **Strategy** video games focus on gameplay requiring careful and skillful thinking and planning in order to achieve victory. In most strategy video games, says Andrew Rollings, "the player is given a godlike view of the game world, indirectly controlling the units under his command.
- 4X refers to a genre of strategy video game with four primary goals: eXplore, eXpand, eXploit and eXterminate. A 4X game can be turn-based or real-time. Perhaps the best known example of this genre is Sid Meier's *Civilization* series.
- The **tactical role-playing game** sub-genre principally refers to games which incorporate gameplay from strategy games as an alternative to traditional RPG systems. Like standard RPGs, the player controls a finite party and battles a similar number of enemies, but this genre incorporates strategic gameplay such as tactical movement on an isometric grid.

Game Taxonomy



- The moniker "**real-time strategy**" (RTS), usually applied only to certain computer strategy games, indicates that the action in the game is continuous, and players will have to make their decisions and actions within the backdrop of a constantly changing game state.
- A **real-time tactics game** shares feature of the simulation and wargame genres, for example Empire Total War. These titles focus on operational aspects and control of warfare. Unlike in real-time strategy games, resource and economical management and building plays no part of the gameplay.
- The term **Turn-based strategy game** (TBS) is usually reserved for certain computer strategy games, to distinguish them from real-time strategy games. A player of a turn-based game is allowed a period of analysis before committing to a game action, and some games allow a certain number of moves or actions to take place in a turn.
- **War games** are a subgenre of strategy video games that emphasize strategic or tactical warfare on a map. Wargames generally take one of four archetypal forms, depending on whether the game is turn-based or real-time and whether the game's focus is upon military strategy or tactics.

Game Taxonomy

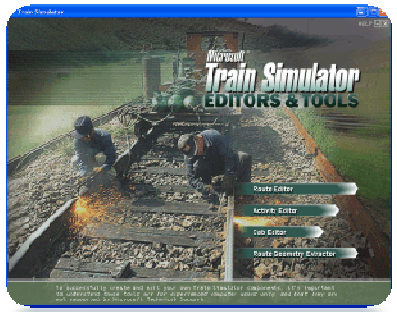




- **Vehicle simulation games** are a genre of video games which attempt to provide the player with a realistic interpretation of operating various kinds of vehicles.
- A **flight simulation** tasks the player with flying an aircraft, usually an airplane, as realistically as possible. Combat flight simulators are the most popular sub-genre of simulation. The player controls the plane, not only simulating the act of flying, but also combat situations. There are also civilian flight simulators that do not have the combat aspect.
- **Racing games** typically place the player in the driver's seat of a high-performance vehicle and require the player to race against other drivers or sometimes just time. This genre of games is one of the staples of the computer gaming world and many of the earliest computer games created were part of this genre.
- **Space flight simulator games** are a sub-genre that involve piloting a spacecraft. Space simulators are different from other sub-genres, and are not generally considered to be simulators, as their simulated objects do not always exist and often disregard the laws of physics.

Game Taxonomy





- **Train simulators** simulate the vehicles, environments and often economics associated with railway transport. These are frequently historical in nature, reminiscing on the evolution and emergence of the railroad in various countries and the economic booms that often accompanied them.
- **Vehicular combat** or *car combat* games focus on fast-paced action, as the player operates a car or other vehicle and attempts to disable or destroy CPU or human opponents. Vehicular combat games often allow a player to choose from a variety of potential vehicles, each with their own strengths and weaknesses.

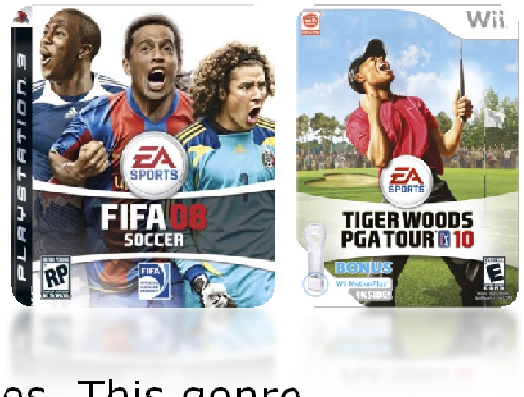
Game Taxonomy





- **Music games** most commonly challenge the player to follow sequences of movement or develop specific rhythms. Some games require the player to input rhythms by stepping with their feet on a dance pad, or using a device similar to a specific musical instrument, like a replica drum set.
- **Party games** are video games developed specifically for multiplayer games between many players. Normally, party games have a variety of different types of mini-games that range between collecting more of a certain item than other players or having the fastest time at something.
- A **programming game** is a computer game where the player has no direct influence on the course of the game. Instead, a computer program or script is written in some domain-specific programming language in order to control the actions of the characters (usually robots, tanks or bacteria, which seek to destroy each other).

Game Taxonomy

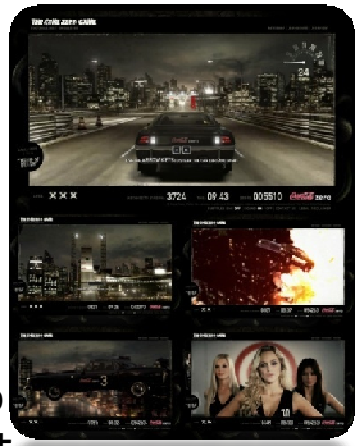


- **Puzzle games** require the player to solve logic puzzles. This genre frequently crosses over with adventure and educational games. Some arcade games, in particular *Tetris*-variants, are often labeled puzzle games, despite the fact that gameplay depends on hand/eye coordination and quick reflexes, rather than thought and logic.
- **Sports games** emulate the playing of traditional physical sports. Some emphasize actually playing the sport, while others emphasize the strategy behind the sport (such as *Championship Manager*). Others satirize the sport for comic effect (such as *Arch Rivals*). One of the best selling series in this genre is the *Madden NFL* series.
- Most popular **board games, card games**, and the like have been computerized to some degree or another. Computer game programs can be worthy opponents and can help improve one's skill at traditional games. Chess, Checkers, Othello (also known as Reversi), and Backgammon have world class computer programs. Mah-jongg and related games are immensely popular in China and Japan

Game Taxonomy

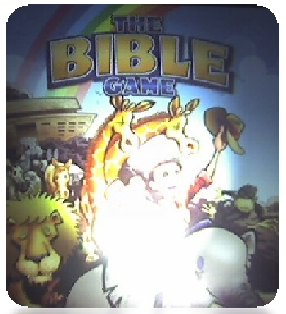
- While most video games are designed as entertainment, many video games are designed with additional purposes. These purposes are as varied as the nature of information itself -- to inform, persuade, or stimulate. These games can have any kind of gameplay, from puzzles to action to adventure.
- **Adult video games**, like adult movies or other media, are intended for an adult audience. In general, the purpose of adult games is to provide erotic entertainment, rather than just gameplay. There exists a wide variety of adult games, though many lack mainstream appeal and represent a niche category. The object of an adult game may differ from a mainstream video game, in that the reward can be a visual representation of nudity, partial nudity, or sexual activity rather than points, etc.

Game Taxonomy



- **Advergames**, in the context of video game genres, refers to promotional software specifically made to advertise a product, organization or viewpoint. The first advergames were distributed on floppy disk by the Chef Boyardee, Coca-Cola, and Samsung brands, while the first cereal box advergame was Chex Quest in 1996.
- **Art games** are designed so as to emphasize art or whose structures are intended to produce some kind of non-ludological reaction in its audience. Art games typically go out of their way to have a unique, unconventional look, often standing out for aesthetic beauty or complexity in design.
- **Casual games** have very simple rules or play techniques and a very low degree of strategy. They also require no long-term time commitment or special skills to play, making them easy to learn and play as a pastime. There are comparatively low production and distribution costs for the producer.

Game Taxonomy



- **Religious games** attempt to provide the dual purposes of spreading a certain religion to non-believers through the medium of video games, and providing gamers with a common pool of games that neither challenge their beliefs nor offend them.
- **Educational games**, as the name implies, attempt to teach the user using the game as a vehicle. Numerous sub-genres exist, each for a different field (math games, typing games and so on).
- **Serious games** are intended to educate or train the player. These games tend to promote "education, science, social change, health care or even the military." Some of these games have no specific ending or goal in the game. Rather, the player learns a real life lesson from the game. For example, games from websites such as Newsgaming.com and gamesforchange.org raise political issues using the distinct properties of games.

Game Taxonomy





- An exergame is a video game that provides exercise. "Exergames" sub-divide into two main implementations, those with a game specifically designed to use an exercise input device (for example, the game *Wii Fit* using the Wii Balance Board) and those implementations using a genre of a game. Games fit into the category of entertainment, and similarly **Exergames** are a category of **Exertainment**. **Exertainment** refers to one aspect of adding entertainment to an exercise workout.

Game Design

- The Art of Computer Game Design, Chris Crowford
  - http://www.vancouver.wsu.edu/fac/peabody/game-book/Coverpage.html#TOC
- **CHOOSE A GOAL AND A TOPIC**
- A game must have a **clearly defined goal**. This goal must be expressed in terms of the effect that it will have on the player. It is not enough to declare that a game will be enjoyable, fun, exciting, or good; the goal must establish the fantasies that the game will support and the types of emotions it will engender in its audience. Since many games are in some way educational, the goal should in such cases establish what the player will learn.
- The importance of a goal does not become obvious until later in the game design cycle. The crucial problems in game development with microcomputers are always **problems of trade-offs**. Everything that the game designer wants to do with her game costs resources, and these are always in short supply with computers. Thus, the designer must make trade-offs. Some game features can be included, and some must be rejected.

Game Design

- The **topic** is the means of expressing the goal, the **environment in which the game will be played**. It is the concrete collection of conditions and events through which the abstract goal will be communicated.
- Most game designers start off by selecting their topic, with their goals subordinated to their topic. Indeed, they commonly describe a game under development by its topic rather than its goal.
- Selecting a good topic can be time-consuming, for each potential topic must be carefully examined for its ability to successfully realize the goals of the game. Many topics carry with them some excess emotional baggage that may interfere with the goals of the game.



Game Design

- **RESEARCH AND PREPARATION**
- Read everything you can on the topic. Study all previous efforts related to either your goal or your topic. What aspects of these earlier efforts appeal to you? What aspects disappoint or anger you? Make sure that you understand the mechanics of the environment your game will attempt to represent.
- During this phase it is critical that you commit little to paper and above all, WRITE NO CODE!
- You will generate during this phase a great variety of **specific implementation ideas** for your game. They will not all fit together neatly, they will require much sorting and rearranging before they can be used. You should not wed yourself to any of them. A large collection of candidates for implementation is a useful resource during the design phase.

Game Design

- **DESIGN PHASE**
- Your primary goal in the design phase is to create the outlines of three interdependent structures: **the I/O structure, the game structure, and the program structure**.
- The **I/O structure** is the system that **communicates information between the computer and the player**.
- The **game structure** is the **internal architecture of causal relationships** that define the obstacles the player must overcome in the course of the game.
- The **program structure** is the organization of **mainline code, subroutines, interrupts, and data** that make up the entire program. All three structures must be created simultaneously, for they must work in concert. Decisions primarily relating to one structure must be checked for their impacts on the other structures.



Game Design

- **I/O Structure**: I/O is composed of input and output. Unlike human languages, the two are not symmetric. The computer has two means of output to the human: graphics on the screen and sound. In the future, we may see more exotic devices for output for games, but for the moment these are the two most common. Graphics are the most important of the two, because we humans are more oriented towards vision than hearing. For this reason, many game designers devote a large portion of their energy towards the design of quality displays. Indeed, some designers go so far as to design the display first and let the game develop from the display, as extreme an example of goal-less design as ever there could be.
- Plan functional, meaningful graphics that convey the critical game information while supporting the fantasy of the game. Don't use graphics tricks as a crutch for a bad game design.
- Use of sound should follow the same rules: use it to tell the player what's going on in the game. The only place where striking but uninformative graphics and sound can be useful is at the beginning of the game, and then only if they help to establish the mood or tone of the game.



Game Design

- **Storyboards** are a graphics design tool that tempt many game designers, for they are a well-developed technology from the film industry. They are not appropriate to games, because storyboards are an intrinsically sequential technology. Games are not sequential, they are branching tree structures.
- The **input structure** is the player's tactile contact with the game; people attach deep significance to touch, so touch must be a rewarding experience for them. The input structure lies at the heart of a fundamental dilemma all game designers must face. An excellent game allows the player to interact heavily with his opponent, to invest a great deal of his personality into the game. This requires that the game offer the player a large number of meaningful options, enough options that the player can express the nuances of his personality through the choices he makes. Yet, decisions must be inputted, and a large number of options seem to require an extensive and complicated input structure, which could well be intimidating to the player. Our dilemma, then, is that an excellent game seems to require a hulking input structure.

Game Design

- The choice of **input device** is an important design decision. I maintain that a good game designer should eschew the use of the keyboard for input and restrict herself to a single simple device, such as a joystick, paddle, or mouse. The value of these devices does not arise from any direct superiority over the keyboard, but rather in the discipline they impose on the designer. Simple input devices go hand-in-hand with simple input structures. Complex input devices encourage complex input structures.
- **Game Structure**: The central problem in designing the game structure is figuring out how to distill the fantasy of the goal and topic into a workable system. The game designer must identify some **key element from the topic environment** and build the game around that key element. This key element must be central to the topic, representative or symbolic of the issues addressed in the game, manipulable, and understandable.

Game Design

- The **key element must be manipulable**, but in a very specific set of ways: that is, it must allow the player to express himself, to do the things that he wants or needs to do to experience the fantasy of the game. For example, in a combat game, shooting is almost always a key element. If the player's freedom to shoot is heavily restricted, the player cannot live the fantasy.
- At the same time, the manipulability must be concise. To use the combat game example again, if the player is required to declare the amount of gunpowder to be expended on each shot, he may well find the manipulability a hindrance to the game.
- Many games employ **multiple key elements**. For example, most combat games include both movement and shooting. This is not necessarily bad; if both key elements are kept simple, or if one key element retains primacy, the game can be successful. However, too many key elements violating too many of these principles will rob the game of its focus.

Game Design

- **Program Structure:** The program structure is the vehicle which translates the I/O structure and game structure into a real product. One of the most important elements of the program structure is the memory map. Definitions of critical variables and subroutines are also necessary. Finally, some documentation on program flow is important.

Game Design

- **EVALUATION OF THE DESIGN**
- The first and most important question is: does this design satisfy my design goals? Does it do what I want it to do? Will the player really experience what I want him to experience? If you are satisfied that the design does pass this crucial test, proceed to the next test.
- Examine the stability of the game structure. Are there any circumstances in which the game could get out of control?
- Now probe the design for unanticipated shortcuts to victory. A player who can find a way to guarantee victory with little effort on his part will not derive the full benefit of your game. Insure that all unintended shortcuts are blocked so that the player must experience those processes that you want him to experience. Any blocks you place must be unobtrusive and reasonable. The player must never notice that he is being shepherded down the primrose path.
- The last and most crucial decision is the decision to abort the game or proceed. Abort if the game no longer excites you. Abort if you have doubts about its likelihood of success. Abort if you are unsure that you can successfully implement it.

Game Design

- **PRE-PROGRAMMING PHASE**
- If the game has made it this far, you are now ready to commit your ideas to paper. Until now your documentation has been sketchy, more along the lines of notes and doodles than documents. Now you are ready to prepare your complete game documentation. First, commit all of your design results from the previous phase to paper. Define the I/O structure and the internal game structure. The tone of this documentation should emphasize the player's experience rather than technical considerations. Compare this first set of documents with your preliminary program structure notes; adjust the program structure documents if necessary.

Game Design

- **PROGRAMMING PHASE**
- Programming itself is straightforward and tedious work, requiring attention to detail more than anything else. Seldom has a game failed solely because the programmer lacked the requisite programming skills. Games have failed to live up to their potential because the programmer did not expend enough effort, or rushed the job, but in few cases has talent or lack of it been the crucial factor in the programming of a game; rather, effort or lack of it is most often the responsible factor. If you place all of your self-respect eggs in the programming basket, I suggest that you get out of game design and work in systems programming. Otherwise, write the code and debug it.

Game Design

- **PLAYTESTING PHASE**
- Playtesting is a process that yields information used to polish and refine the game design. In practice, playtesting often reveals fundamental design and programming problems that require major efforts to correct. Thus, playtesting is often interwoven with a certain amount of program debugging.
- Sometimes playtesting reveals that the game is too seriously flawed to save. A nonfatal, correctable flaw is usually a matter of insufficiency or excess: not enough color, too many pieces, not enough action, too much computation required of the player. A fatal flaw arises from a fundamental conflict between two important elements of the game whose incompatibility was not foreseen. You must have the courage to trash a game with such a fatal flaw. Patching after the game is programmed can only achieve limited gains; if the game is badly deformed, abortion is preferable to surgery.

Game Design

- If your initial design was well-developed (or you are just plain lucky) the game will not face such crises; instead, the problems you will face will be problems of polish. All of the little things that make a game go will be out of tune, and the game will move like a drunken dinosaur instead of the lithe leopard you had envisioned. **Tuning the game** will take many weeks of work. For the short term you can scrimp on the tuning while you are working on other problems, for tuning the game requires delicate adjustments of all the game factors; any other changes will only throw off the tune. Therefore, defer final tuning work until the very end of the polishing stage.
- There are actually two forms of **playtesting**. The first is your **own playtesting** done in the final stages of debugging. The second form comes later when you **turn over the game to other playtesters**. The salient-difference between the two lies in the nature of the bugs exposed. Your own playtesting should reveal and eliminate all program bugs (arising from flaws in the program structure) and many of the game bugs (arising from flaws in the game structure).

Game Design

- **The game you give to the playtesters should be free of program bugs**; they should discover only bugs in the game structure. There is no point in showing an incomplete game to playtesters, and indeed there is a danger in contaminating their objectivity by showing them a version of the game too early.
- Playtesters must be selected and used with great care. You need playtesters who possess a deep familiarity with games, playtesters who can analyze and criticize your game with some basis of experience. You should also know the player well, both his personality and his game taste. You should never use more than five or six playtesters.
- **Schedule along interview with the playtester after he has had enough time to digest the game**. Come to the interview prepared with a set of standard questions that you ask all playtesters. Do not lead the playtester's answers and don't solicit praise. Your job is to find flaws; accolades come later.

Game Design

- The final stage of the design cycle is devoted to **polishing the game**. The polishing stage is actually concurrent with the later stages of playtesting and may involve several iterations with the playtesters.
- This stage is critical; the designer has been working on the game for a long time by now and the luster of the new design has worn off. It is now only a big job that should have been finished months ago. The playtesters love it, the publisher loves it and wants it right now, and the designer is sick of it.
- One of the last tasks you must perform before releasing the game is **the preparation of a game manual**. Manuals are frequently given short shrift by just about everybody associated with computer games. This is a serious mistake, for the manual is a vital element in the overall game package. A computer has many limitations; some can be overcome with a good manual. Much of the static information associated with a game can be presented in a manual. The manual is also an excellent place to add fantasy support elements such as pictures and background stories. Finally, a well-written manual will clear up many of the misunderstandings that often arise during a game.



Game Design

- **POST-MORTEM**
- Once the program is out, brace yourself for the **critics**. They will get their filthy hands on your lovely game and do the most terrible things to it. They will play it without reading the rules. If it's a strategic game, they will castigate it for being insufficiently exciting; if it's an S&A game, they will find it intellectually deficient. They will divine imaginary technical flaws and speculate incorrectly on your deep psychological hang-ups that led you to produce such a game.
- The **public** is another matter. If they don't buy your game, you lose two ways: first, you or your employer make little money on the game; and second, you don't reach as many people with your message. It doesn't matter how beautiful your message is-if nobody listens to it, you have failed as an artist. One failure is nothing to worry about; every artist bombs occasionally. Two failures in a row are bad; three should initiate a serious reconsideration of artistic values

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